02.02.2024

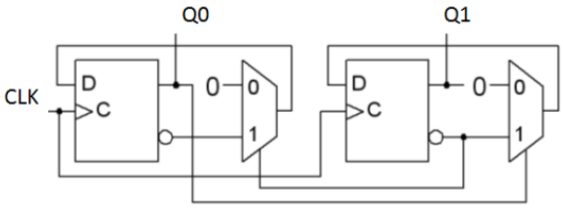
EE2003 and CSE 3017 DIGITAL DESIGN FINAL MAKE-UP EXAM

Name:

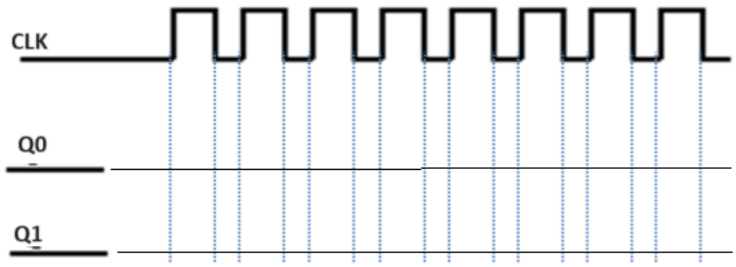
DURATION: 90 MINUTES
CLOSED BOOK, CLOSED NOTES EXAM. NO CALCULATORS OR CHEAT SHEETS ALLOWED.
PLEASE PROVIDE YOUR ANSWERS IN DESIGNATED PLACES.

Q1) For the sequential circuit below,

(A) **(8 points)** Write state equations.

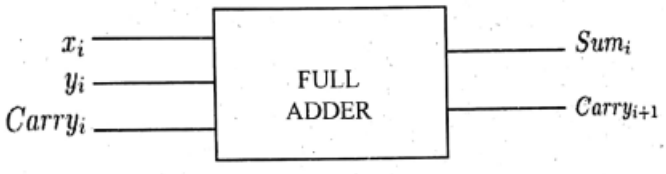


(B) **(8 points)** If the circuit starts in the state $Q_1Q_0 = 00$, sketch output states.



(C) **(4 points)** What does this circuit do?

Q2) A full adder circuit using has a general block diagram as below.



(A) **(8 points)** Form truth table. (x and y are single bit variables)

(B) **(10 points)** Implement the circuit using a 3x8 decoder and appropriate logic gates.

Q3) Design a combinational logic circuit which receives a 4-bit unsigned number X (x_3 x_2 x_1 x_0) as input and produces an output Z which equals the result of integer division of X by 3 (e.g., if X=7, Z=2).
(A) **(8 points)** Form the truth table.

(B) **(10 points)** Determine the minimized output functions in SoP form using K-maps.

Q4) (10 points) For a RAM application (with depth = 4 and width = 8), an excerpt of testbench file is given below. Write main module.

```
ram UUT(.clock(clock), .write_enable(write_enable),
        .address(address),
        .data_in(data_in), .data_out(data_out));

always #2 clock = ~clock;

integer i;

initial begin
    clock = 0;
    write_enable = 0;
    address = 4'h0;
    data_in = 8'h00;
    #9 write_enable = 1;

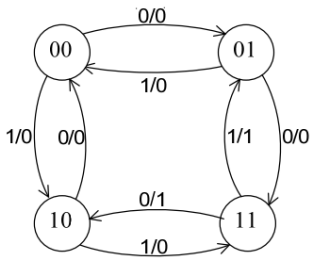
    for(i = 0; i < 16; i = i + 1) begin
        address = i;
        data_in = i * 16;
        #4;
    end

    write_enable = 0;

    for(i = 0; i < 16; i = i + 1) begin
        address = i;
        #4;
    end

end
```

Q5) Given the state diagram below,
(A) **(6 points)** Form state table.



(B) **(10 points)** Write Verilog description using case statements.

(C) **(8 points)** Minimize the outputs in SoP form using K-maps.

(D) **(10 points)** Implement the sequential circuit using D flip flops.